

DeGro: Target Determinacy Verification and Grounded Repair for Executable Mathematical Reasoning

Anonymous Author(s)
Submission Id: ANONYMOUS

Abstract

Executable mathematical reasoning may run successfully while solving an underconstrained or mistranslated problem. Execution checks expose syntax, runtime, and arithmetic failures but do not establish that the encoded constraints determine the requested quantity. We introduce *DeGro*, a neuro-symbolic framework for *target determinacy diagnosis*. DeGro asks a solver for two feasible assignments that disagree on the target; when they exist, a repair policy requests source support and accepts a constraint only if it passes provenance and formal checks that restore determinacy; otherwise, it abstains. On a 300-pair MIRA-Math evaluation that varies source support while holding the formalization constant, DeGro improves repair-or-abstain accuracy for GPT-OSS 20B and 120B. Human-verified natural-error studies show that DeGro improves accuracy on the supported MATH-500 cohort, while higher-coverage ALG514 reveals answer-compatible formalization errors. Detection is much easier than repair, and concrete witnesses add no measurable benefit beyond the verdict. Target determinacy is therefore an auditable intermediate guardrail, not a proof of source-to-formal faithfulness. An additional 300-pair, author-reviewed DRAW-Paired experiment uses natural algebra sources with injected omissions: DeGro improves FDA over grounded self-review by 8.8 points for GPT-OSS 120B and 10.2 for Nano, and 5.8 for 20B; DeGro ties verdict-only repair for the highest 120B FDA. This subset analysis is post-hoc: review and the subsequent 50-pair exclusion followed inference, and the extension was selected after initial outcomes.

CCS Concepts

• Computing methodologies → Knowledge representation and reasoning; Intelligent agents; Natural language processing.

Keywords

neuro-symbolic reasoning, mathematical agents, verification, counterexamples, abstention

ACM Reference Format:

Anonymous Author(s). 2027. DeGro: Target Determinacy Verification and Grounded Repair for Executable Mathematical Reasoning. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, 3–7 May 2027, IFAAMAS, 12 pages.



This work is licensed under a Creative Commons Attribution International 4.0 License.

Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027), M. Baldoni, F. Fang, W. Yeoh, N. Yorke-Smith (eds.), 3–7 May 2027, Hanoi, Vietnam. © 2027 International Foundation for Autonomous Agents and Multiagent Systems (www.ifaamas.org).

1 Introduction

Large language models increasingly delegate mathematical computation to programs and symbolic solvers. Program-aided reasoning reduces arithmetic errors, while systems such as SymCode generate self-contained symbolic programs and repair execution failures [5, 15]. Successful execution, however, does not establish that the encoded constraints determine the requested answer. A program may run and return a value even when its formalization admits several values of the target.

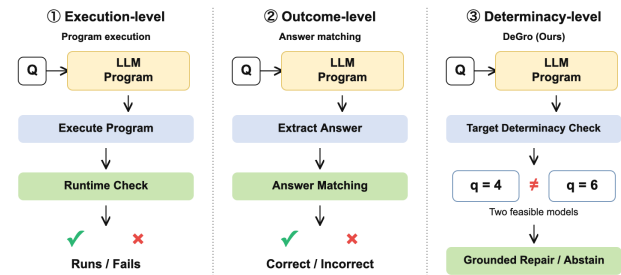


Figure 1: Three levels of executable mathematical reasoning evaluation. Execution checks whether a program runs; outcome evaluation checks its final answer. DeGro diagnoses target ambiguity through two feasible models with different target values (illustrated by $q = 4$ and $q = 6$), then attempts source-grounded repair with deterministic acceptance gates or abstains. Target determinacy is relative to the encoded specification and does not establish faithfulness to the source problem.

Consider: “Ten tasks are divided between A and B. A receives two more than B. How many does A receive?” A generated model containing only $a + b = 10$ is executable and satisfiable, but $(a, b) = (4, 6)$ and $(6, 4)$ imply different values of the target a . We call this condition *target ambiguity*: feasible assignments disagree on the requested quantity. It concerns the target rather than uniqueness of the entire assignment, since freedom in other variables need not affect the answer.

Detecting ambiguity does not establish how to resolve it. The same model faithfully represents the shorter source “Ten tasks are divided between A and B. How many does A receive?” For the complete source, adding $a = b + 2$ recovers an omitted relation; for the shorter source, the same addition fabricates information. The formalization and solver verdict are identical, but the correct action changes with source support. A repair policy must therefore distinguish evidence that the target is ambiguous from evidence that a particular constraint is justified.

We propose DeGro, a neuro-symbolic framework that separates these two decisions. It first constructs two renamed copies of a generated specification and asks whether both can satisfy the constraints while disagreeing on the target. A satisfying query provides auditable evidence of ambiguity. The agent then proposes a constraint with supporting source text, and acceptance checks require consistency, non-redundancy, and restored target determinacy. If no admissible repair is found, the agent abstains. The solver guarantee concerns the encoded specification: even a determinate target can be wrong when the target or constraints were mistranslated.

We evaluate this framework at two complementary levels. A controlled paired benchmark holds the formalization and diagnostic fixed while varying source support, testing whether the policy repairs a supported omission and abstains on its underspecified counterpart. Human-verified natural-error studies then examine representation coverage, ambiguity detection, and successful grounded repair separately. The controlled results show improved repair-or-abstain decisions, while the natural errors expose a substantial gap between detecting ambiguity and repairing its cause. We also test whether concrete witness assignments improve decisions beyond the solver verdict.

Our research questions are:

(RQ1) Decision quality and tradeoffs. How do target-determinacy diagnosis and source grounding affect repair-or-abstain accuracy and the tradeoff between supported repair and unsupported additions?

(RQ2) Paired source sensitivity. When the formalization and diagnostic are held fixed, can an agent both repair the source-supported omission and abstain on the underspecified member of the same pair?

(RQ3) Practical bottlenecks. How do representation coverage, ambiguity detection, and grounded repair limit effectiveness on natural errors, and do concrete witnesses improve controlled repair decisions beyond the verdict?

Takeaway

DeGro checks whether the encoded constraints determine the requested target, requires cited support and formal validation for repairs, and otherwise abstains.

Our main contributions are:

(1) A target-determinacy-guided repair-or-abstain framework. We introduce DeGro, which combines two-copy target-determinacy diagnosis with source-grounded repair, deterministic acceptance gates, and explicit abstention (RQ1).

(2) Controlled and natural-source paired benchmarks. We construct matched cases that hold the formalization fixed while varying whether the source supports the missing constraint. Alongside MIRA-Math, DRAW-Paired contributes 300 retained author-reviewed pairs derived from natural algebra word problems, isolating supported repair from unsupported completion (RQ2).

(3) A multi-stage empirical audit of executable reasoning. We combine controlled and human-verified natural-error evaluations to separate representation coverage, ambiguity detection, and grounded repair; we additionally test whether concrete witnesses improve decisions (RQ3).

The formal guarantee concerns F ; the empirical answers concern model and policy behavior.

2 Related Work

2.1 Executable Mathematical Reasoning

PAL and Program-of-Thoughts separate language-based reasoning from program execution to reduce computational errors [8, 15]. MathCoder and ToRA integrate code and tool calls into mathematical problem solving [18, 35], while faithful chain-of-thought connects generated rationales to executable symbolic procedures [25]. SymCode is particularly close to our setting: it represents mathematical reasoning as a SymPy program and uses execution feedback to repair programmatic failures [5]. These approaches motivate an additional check after successful execution. A generated artifact may return a value even though its constraints admit other values of the requested quantity. DeGro checks this property of the encoded feasible region and routes ambiguous specifications to grounded repair or abstention.

Outcome verification and process supervision assess candidate solutions or intermediate reasoning steps [10, 24, 33]. Self-Refine uses model-generated feedback, and CRITIC incorporates external tools into revision [17, 26]; evidence on intrinsic self-correction shows the limits of revision without reliable external feedback [20]. DeGro contributes a specific solver observation: two feasible assignments disagree on the target. This observation establishes ambiguity under the representation, independently of the predicted answer, but does not identify which source constraint should be added.

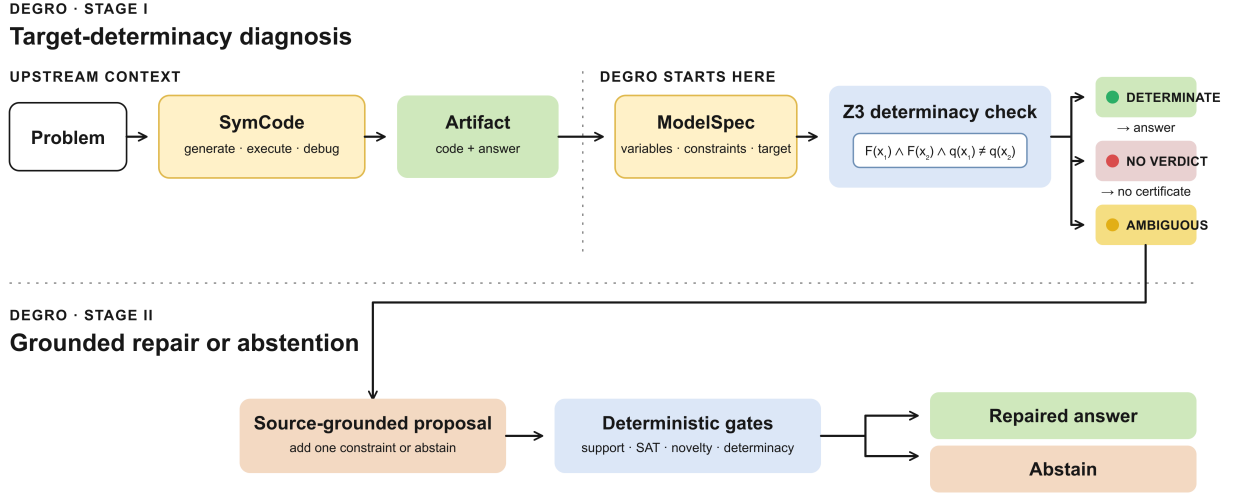


Figure 2: DeGro in the executable-reasoning pipeline. SymCode generates the executable artifact, while DeGro passes a structured MODELSPec to Z3-backed target-determinacy verification. A determinate target can be answered, an ambiguous target enters grounded repair and validation, and an inconclusive check returns no certificate.

2.2 Symbolic Diagnosis and Relational Verification

Recent work addresses semantic failures that survive execution. ReLoop tests optimization formulations through solver-based parameter perturbations [22]. SymDiag translates reasoning steps into constraints and uses satisfiability and entailment checks to localize defects, with a Self-Auditor to distinguish translation errors from reasoning errors [11]. Minimal-core-guided repair instead returns an unsatisfiable subset of generated constraints as evidence for revising inconsistent translations [23, 30]. DeGro focuses on a complementary condition: the specification is satisfiable, but its feasible assignments disagree on the requested target. Its diagnosis concerns the specification as a whole rather than the validity of individual derivation steps, and its additive repair policy requires support from the original source.

The two-copy construction builds on self-composition and product programs, which reduce relational verification to queries over renamed copies [6, 7, 32]. DeGro instantiates this established technique as target agreement across feasible assignments. Agreement on the target is weaker than uniqueness of the full assignment: nuisance variables may remain free without making the answer ambiguous. Counterexample-guided refinement and synthesis likewise use solver evidence to guide changes [9, 31]. Here, however, a counterexample refutes target determinacy without supplying a semantically justified edit. Our contribution is the integration of this diagnosis with a source-sensitive repair policy and its evaluation, rather than a new relational verification primitive.

2.3 Grounded Repair, Abstention, and Faithfulness

Classical diagnosis, abductive inference, and formal model repair provide foundations for identifying defects and modifying specifications to satisfy a desired property [3, 13, 29]. LLM-based repair extends this direction to declarative formal specifications [2]. Restoring determinacy alone is insufficient in our setting, since a guessed equality can force a unique target. DeGro therefore requests a source-supported constraint and checks consistency, non-redundancy, and restored determinacy before acceptance. These formal gates verify the effect of the edit; source support remains a separate semantic requirement.

Selective prediction studies when to withhold an answer under a risk-coverage tradeoff [14, 16], and semantic uncertainty estimates uncertainty in language-model outputs [21]. DeGro bases its repair-or-abstain decision on two distinct questions: whether the encoded target is determinate and whether the source supports a constraint that resolves ambiguity. MIRA-Math is directly relevant because it evaluates requesting and integrating necessary missing information [1]. We adapt its validated latent states into matched sources with identical formalizations but different support for repair, isolating the decision to recover an omitted relation or abstain from unsupported completion.

Autoformalization also separates formal validity from preservation of natural-language meaning [4, 36]. DeGro’s guarantee is similarly representation-relative: a determinate specification can still contain a wrong target or an unsupported constraint. Our paired and human-verified natural-error evaluations therefore distinguish determinacy diagnosis, grounded repair, and answer accuracy rather than treating a solver verdict as proof of source-to-formal faithfulness.

3 Methodology

DeGro checks whether a generated mathematical specification determines the requested target. Stage I diagnoses target ambiguity using a symbolic solver. Stage II attempts to resolve an ambiguous target by adding a constraint with source provenance, or abstains if no candidate is accepted. Figure 2 summarizes the pipeline.

3.1 Problem Setup and Target Determinacy

Given a natural-language problem P , the model produces a specification $F = (V, D, C, q)$, where V is the set of variables, D their domains, C the constraints, and $q(V)$ the requested expression. Its feasible assignments are

$$\text{Sol}(F) = \{m \mid m \models D \wedge C\}. \quad (1)$$

For a satisfiable F , the target is *determinate* when every feasible assignment gives the same target value:

$$\text{Det}(F, q) \iff \forall m_1, m_2 \in \text{Sol}(F), \quad q(m_1) = q(m_2). \quad (2)$$

Target determinacy does not require a unique assignment to all variables. For example, $x = 5, y \geq 0$ fixes the target $q = x$ even though y remains free. An inconsistent specification is reported separately and never receives a determinacy certificate.

The specification is stored as a typed `MODEL SPEC`: a structured record of variable sorts and bounds, source-linked constraints, and the target. Computational and verification backends consume this same representation, avoiding discrepancies between independently generated programs. The supported fragment includes integer, real, and Boolean variables, arithmetic equalities and inequalities, finite domains, and selected modular constraints.

The certificate concerns F , not whether F correctly represents P . Wrong or extra constraints can fix a target while mistranslating the source. DeGro therefore repairs only ambiguous specifications by adding constraints; correcting a determinate but incorrect representation is outside its scope.

3.2 Stage I: Target-Determinacy Diagnosis

DeGro first checks whether $D \wedge C$ is satisfiable. An unsatisfiable result yields `INCONSISTENT`; a timeout or inconclusive solver result yields `UNKNOWN`. If the specification is satisfiable, DeGro constructs two copies with disjoint variable names and asks whether they can produce different target values:

$$\Phi(F, q) = F^{(1)} \wedge F^{(2)} \wedge q(V^{(1)}) \neq q(V^{(2)}). \quad (3)$$

If Φ is satisfiable, its projections form a *target-divergent witness pair*, and the status is `AMBIGUOUS`. If Φ is unsatisfiable, the status is `DETERMINATE`, since the initial check has already established that F has a solution. Unsupported representations yield `UNSUPPORTED` before solver verification. Only `AMBIGUOUS` enters Stage II.

Algorithm 1 Target-determinacy verification

Require: Typed specification $F = (V, D, C, q)$

Ensure: Status and optional witness pair

```

1:  $r_0 \leftarrow \text{CHECKSAT}(D \wedge C)$ 
2: if  $r_0 = \text{UNSAT}$  return INCONSISTENT
3: if  $r_0 \neq \text{SAT}$  return UNKNOWN
4:  $\Phi \leftarrow F^{(1)} \wedge F^{(2)} \wedge (q^{(1)} \neq q^{(2)})$  using disjoint copies
5:  $(r_1, M) \leftarrow \text{CHECKSAT}(\Phi)$ 
6: if  $r_1 = \text{UNSAT}$  return DETERMINATE
7: if  $r_1 \neq \text{SAT}$  return UNKNOWN
8: return AMBIGUOUS, PROJECTWITNESSES( $M$ )

```

Soundness. Assuming semantics-preserving compilation and a sound solver for the supported theory, $\text{SAT}(F)$ together with $\text{UNSAT}(\Phi)$ establishes target determinacy: any two feasible assignments with different targets would satisfy Eq. (3), contradicting unsatisfiability. Conversely, a satisfying assignment to Φ supplies two feasible witnesses with different targets. Timeout and `UNKNOWN` provide no certificate. The implementation uses a restricted expression compiler and separate variable maps for the two Z3 copies [12].

3.3 Stage II: Grounded Repair or Abstention

The verifier retains the target-divergent witness pair as a certificate, but the default repair prompt receives only the source problem, the encoded constraints and target, and the typed `AMBIGUOUS` verdict. It returns either `ADD-CONSTRAINT` with an exact supporting source span (or a permitted domain rule), or `ABSTAIN`. It receives neither the hidden gold target nor the missing gold constraint. An exploratory ablation additionally exposes the concrete witnesses to test whether their values help beyond the verdict; they do not improve decisions on the present benchmark. In either variant, the model must recover a relation from the source rather than select a witness as correct.

Acceptance gate. For a proposed constraint c and a frozen set $\mathcal{R}$ of permitted domain-semantics rules, the deterministic gate accepts c only when

$$\begin{aligned} \text{Accept}(c) \iff & \text{Compiles}(c) \wedge \text{Provenance}(c, P, \mathcal{R}) \\ & \wedge \text{SAT}(F \wedge c) \wedge \neg(F \models c) \wedge \text{Det}(F \wedge c, q). \end{aligned} \quad (4)$$

These checks require that the candidate compile, cite a source span occurring in P or a rule in $\mathcal{R}$, preserve consistency, add a non-redundant restriction, and restore target determinacy. Compilation failures and inconclusive checks do not permit acceptance.

Provenance is an auditable check, not a proof that the constraint correctly expresses the cited text. Passing the gate therefore establishes valid provenance and determinacy under the repaired specification, rather than semantic correctness with respect to P . Semantic source support and functional correctness are evaluated separately for repair credit; an incorrect translation or unsupported target pin does not earn credit merely by passing the gate.

Bounded repair. DeGro allows at most two proposal attempts. The first accepted constraint is committed and terminates the loop. If the first proposal is rejected, the second attempt receives the rejection reason and operates on the unchanged F . Rejected constraints are not accumulated: each candidate must restore determinacy on its own, and a supported but insufficient addition is rejected. If

Table 1: Controller semantics for DeGro integration.

Verifier state	Evidence	Controller action
Determinate	F sat, Φ unsat	answer with scope
Ambiguous	targets disagree	grounded repair
Inconsistent	F unsat	revise encoding
Unknown	solver inconclusive	fallback/escalate
Unsupported	outside fragment	alternate tool
No repair found	no accepted candidate	ask or abstain

the model abstains or neither candidate passes, the outcome is NO-ADMISSIBLE-REPAIR-FOUND, and F is preserved.

Failure to find an admissible repair does not prove that the source is underspecified: the model may have missed a relation already present in the source.

Controller policy. Table 1 maps verifier states to agent actions. A determinate result permits an answer scoped to the current specification, including an accepted repair when present. In interactive use, failure to find a repair can trigger clarification or abstention; the benchmark scores abstention because no clarification channel is available. Inconsistent, unknown, and unsupported states require distinct fallbacks rather than being treated as ambiguous inputs. The parser, compiler, solver, and gate form the trusted boundary; the language model only proposes specifications and repairs. Logs retain solver verdicts, witnesses, proposals, cited support, and rejection reasons for auditing.

4 Experiments and Results

4.1 Setup

Compared systems and model settings. To answer RQ1, we compare SELF-REVIEW, GROUNDED SELF-REVIEW, DETERMINACY-GUIDED REPAIR, and DeGro under the same initial formalization and a two-attempt repair budget. All methods receive the source, current constraints, and target, ensuring that performance differences reflect repair behavior rather than initial-generation quality.

Self-review asks the model to revise the formalization without an explicit diagnostic signal or grounding requirement. Grounded self-review adds the same exact-span grounding rule used by DeGro, requiring every introduced constraint to be supported by the source, but receives neither the solver verdict nor witness information. Determinacy-guided repair receives the typed AMBIGUOUS verdict without the grounding requirement, isolating the effect of determinacy feedback. DeGro combines the ambiguity verdict with exact-span grounding and applies the same deterministic acceptance gate used during evaluation. In the main comparison, witness values are not exposed to the repair model.

We evaluate these methods with GPT-OSS 20B and GPT-OSS 120B [28], and Nemotron 3 Nano 30B [27], using the same cases, methods, frozen prompts, temperature 1.0, and matched medium-reasoning API settings, yielding $300 \times 2 \times 4 = 2,400$ complete records per model. Because identical API settings do not necessarily correspond to identical internal reasoning budgets across model families, cross-model comparisons are interpreted as robustness checks rather than strictly compute-matched comparisons.

For RQ2, the paired construction evaluates whether repair behavior is sensitive to evidence in the source. For the controlled component of RQ3, we use a separate exploratory development set of 100 pairs to test whether the repair model benefits from seeing two concrete assignments that both satisfy the current formalization but yield different target values, rather than receiving only the typed AMBIGUOUS verdict.

4.2 Data

Controlled paired benchmark (RQ2). For each gold specification $F^* = \{c_1, \dots, c_k, c^*\}$, we form $F^- = F^* \setminus \{c^*\}$ and retain it only if F^* determines the target while F^- is satisfiable and target-ambiguous. Each pair shares this same underconstrained formalization F^- . In the OMISSION case, the source still states the fact represented by c^* , so the formalization has omitted source information and repair is correct. In the UNDERSPECIFIED case, that fact is absent from the source itself, so no source-supported constraint can restore it and abstention is correct. The two cases therefore have the same constraints and solver verdict but require opposite actions; the source is the only label-revealing input. We report all 300 MIRA-Math pairs (600 cases, five families) [1]: 80 pilot, 100 development, and 120 held-out pairs. Because pilot and development outcomes informed method development, this is not an independent held-out test. The supplement reports the 120-pair subset separately.

Integrity checks. The builder imports the typed missing fact from MIRA-Math rather than asking an LLM to corrupt the problem, then applies all three solver checks. The 300-pair set exhausts compatible source identifiers; its manifest records source revision 72ff918, license, selection seed, family counts, and prompt/output hashes. The five families cover linear separation, rank-deficient systems, path sums, Chinese-remainder reconstruction, and moment constraints.

The families exercise distinct structures. Linear separation removes a relation that distinguishes otherwise compatible solutions; rank-deficient systems preserve several degrees of freedom. Graph path sums and Chinese-remainder reconstruction combine discrete equalities with domain restrictions, while moment problems couple polynomial summaries within the supported degree bound. Family-stratified results check that one easy template does not dominate the aggregate. Because every retained deletion is mechanically verified as target-critical, the benchmark does not mislabel an irrelevant omission as a repair opportunity.

Main natural-source paired benchmark (DRAW-Paired). DRAW-Paired uses DRAW-1K’s 600 training and 200 development problems [34]; its 200 test problems remain untouched. Excluding one duplicate identifier and 200 single-equation problems leaves 599 solver-eligible systems: deleting a gold equation preserves satisfiability but makes a real-valued target ambiguous, while the full-system target matches a published gold solution. Gold coefficient alignments propose supporting sentences. The initial seeded cohort has 300 pairs (223 training-derived, 77 development-derived). A source-disjoint, split-proportional extension selects 50 of 67 remaining unflagged proposals (37 training-derived, 13 development-derived), giving an initial cumulative pool of 350 pairs and 700 cases. A subsequent author re-review excludes 50 pairs (44 initial, six extension), leaving 300 pairs and 600 cases from 221 training

and 79 development sources. Exclusions followed inference and access to model outcomes; the released exclusion list records all removed identifiers. Pair members share retained equations, target, and domains; only the supporting span is deleted in the underspecified member. These are injected omissions on natural sources, not naturally occurring model errors.

All retained spans and deletions were author-reviewed after inference; the subsequent 50-pair exclusion was also post-hoc. Two initial title-abbreviation errors were corrected and affected pairs rerun; other responses were retained and rescored. The extension was selected after initial outcomes, locked before its inference, then approved without input changes. Thus this human-reviewed analysis is post-hoc, not preregistered or human-frozen before inference. Each model has 2,400 retained records with shared policies and decoding settings; excluded responses are removed from released raw and complete outputs. Retries retain the first valid response without score-based resampling. DeGro-minus-grounded-self-review FDA intervals use 10,000 pair-cluster bootstrap resamples; Holm correction covers three models. The supplement details provenance.

Natural-error study. For ecological validity, GPT-OSS 20B generated one Structured SymCode specification per MATH-500 problem without injected errors. We froze all parsed, solver-supported, conclusive cases before annotation or outcome inspection and reused each initial specification across methods. Model-proposed labels and evidence were hidden from verifier outputs, then fully checked by two reviewers: all 292 MATH-500 and 503 ALG514 cases. Because separate pre-discussion human labels were not retained, we claim human verification, not independent annotation or inter-annotator agreement. Labels distinguish correct, missing/wrong/extra constraint, wrong domain, wrong target, computational error, and other; only source-supported target-critical omissions enter the primary subset.

For MATH-500, two reviewers inspected all 292 final labels from an archived model-assisted review. GPT-OSS 120B subsequently produced a fresh blinded proposal for every case under the same prompt; it agreed with the human-verified primary label on 207 of 292 cases (70.9%), illustrating why model proposals were not treated as ground truth. For ALG514, GPT-OSS 120B proposed 503 labels and re-adjudicated all 22 cases where semantic status disagreed with verifier-compatible numerical status; reviewers then inspected all final labels. Detector ambiguity could not define the positive class: a positive required a source-supported omission that can change the requested target.

For RQ3, primary natural-error measures are ambiguity recall and end-to-end grounded repair, with conclusive solver coverage reported over all inputs; secondary measures include precision, false ambiguity/repair, and preservation of correct controls. Proportions use Wilson 95% intervals and paired answer accuracy uses exact McNemar. Numeric agreement alone is not repair success.

The original problem is the unit of analysis, and every method reuses its shared initial formalization. Recall asks how often DeGro identifies verified target-critical omissions; precision asks how often an ambiguous verdict corresponds to that class. Conditional repair counts accepted grounded fixes among detected omissions, whereas end-to-end repair retains every verified omission in the

denominator. This distinction exposes the gap between finding ambiguity and recovering the missing source relation.

MATH-500 contributes 292 of 500 conclusive, solver-supported formalizations; ALG514 contributes 503 of 514. Exclusions are reported in the coverage results. The original DRAW-1K natural-error audit remains omitted because corrected-verifier outcomes and final human labels were incomplete; this is separate from the gold-equation-derived DRAW-Paired experiment.

4.3 Main Paired Experiment on MIRA-Math

Primary comparison. We compare DeGro with grounded self-review on FDA using exact paired McNemar tests and Holm correction across three models. Percentage-point differences use 10,000 pair-cluster bootstrap resamples. The 300-pair comparison includes method-development observations; the supplement retains the 120-pair held-out results and original analysis protocol. GPT-OSS supports configurable reasoning effort and structured outputs [28].

Metrics and analysis. For RQ1, the primary metric is final decision accuracy (FDA): a case is correct only when the method makes a grounded, functionally correct repair on an omission case or abstains on an underspecified case. Because every controlled pair contains one case of each type, $FDA = (RSR + CAR)/2$. Repair success rate (RSR) is the proportion of omission cases repaired correctly, and correct abstention rate (CAR) is the proportion of underspecified cases on which the method abstains. Unsupported repair (UR) is the share of proposed additions that lack source support; it must be interpreted alongside RSR because always abstaining would trivially produce no unsupported additions. Over-repair rate equals $1 - CAR$ and is omitted from the main table as redundant. Pair-cluster confidence intervals preserve within-pair dependence; semantic, functional, and grounded outcomes remain separate.

4.4 Main Results

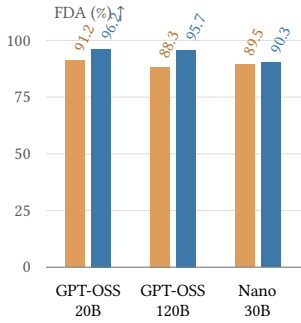
RQ1: repair-or-abstain decisions. Table 2 reports all 300 MIRA-Math pairs. DeGro raises FDA from 91.2% to 96.2% for GPT-OSS 20B (+5.0 points, 95% CI [2.3, 7.8]; 49 versus 19 discordances; exact $p = .000358$) and from 88.3% to 95.7% for 120B (+7.33 points, CI [4.7, 10.0]; 57 versus 13; $p = 1.03 \times 10^{-7}$). Nano changes from 89.5% to 90.3% (+0.83 points, CI [-1.5, 3.3]; 30 versus 25; $p = .590$), an inconclusive difference. Holm-adjusted p -values are .000716, 3.09×10^{-7} , and .590. DeGro has the highest FDA among the four policies for each model, without a demonstrated incremental gain for Nano.

RQ1: source-grounded acceptance. On the 300 pairs, 20B DeGro reduces UR from 12.5% to 3.4% and raises CAR from 86.7% to 99.0%, while RSR falls from 95.7% to 93.3%. For 120B, UR falls from 17.3% to 7.0%, CAR rises from 79.7% to 93.3%, and RSR from 97.0% to 98.0%. Nano reaches 100.0% CAR and 0.8% UR, versus 99.0% and 2.8%, with similar RSR (80.7% versus 80.0%). These metrics expose model-specific repair-abstention tradeoffs. Figure 4 illustrates rejection of a superficially grounded but non-resolving edit.

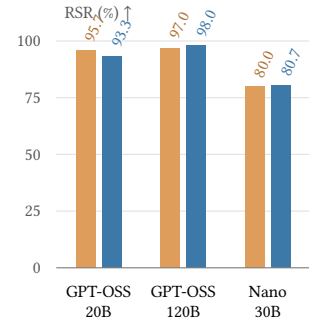
Table 2: Main paired experiments on MIRA-Math (300 pairs, 600 cases) and DRAW-Paired (300 retained author-reviewed pairs, 600 cases). Values are percentages. FDA is primary; RSR and CAR decompose repair and abstention, and UR measures unsupported additions among proposed additions. Blue rows denote DeGro; bold FDA marks the best policy within each model and dataset. MIRA-Math includes pilot and development observations; its split protocol is described in the methods. DRAW-Paired review followed inference, and its extension was selected after initial outcomes.

Model	Method	MIRA-Math: controlled sources				DRAW-Paired: natural sources			
		FDA↑	RSR↑	CAR↑	UR↓	FDA↑	RSR↑	CAR↑	UR↓
GPT-OSS 20B	Self-review	89.7	92.3	87.0	15.5	73.8	66.7	81.0	43.3
	Grounded self-review	91.2	95.7	86.7	12.5	73.3	62.3	84.3	43.8
	Determinacy-guided	92.8	87.7	98.0	6.7	73.7	57.0	90.3	46.1
	DeGro (ours)	96.2	93.3	99.0	3.4	79.2	66.7	91.7	36.3
GPT-OSS 120B	Self-review	77.7	100.0	55.3	30.9	72.0	68.3	75.7	44.1
	Grounded self-review	88.3	97.0	79.7	17.3	69.0	54.0	84.0	51.1
	Determinacy-guided	92.5	99.0	86.0	13.2	77.8	69.3	86.3	37.5
	DeGro (ours)	95.7	98.0	93.3	7.0	77.8	63.7	92.0	38.8
Nemotron	Self-review	84.5	76.7	92.3	25.3	63.5	44.7	82.3	59.0
Nano 30B	Grounded self-review	89.5	80.0	99.0	2.8	60.7	35.7	85.7	61.9
	Determinacy-guided	73.3	51.3	95.3	24.9	63.3	34.7	92.0	55.4
	DeGro (ours)	90.3	80.7	100.0	0.8	70.8	48.3	93.3	44.4

(a) Final decision accuracy



(b) Grounded repair success



Grounded self-review DeGro

(c) Correct abstention

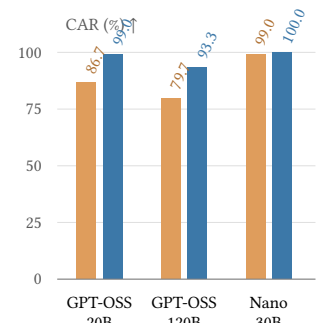


Figure 3: MIRA-Math results on 300 pairs. DeGro has higher FDA than grounded self-review for all three models; the Nano difference is statistically inconclusive. RSR and CAR expose model-specific tradeoffs. All axes start at zero; these results include the pilot and development subsets.

4.5 Main Natural-Source Experiment: DRAW-Paired

Table 2’s right-hand block reports the retained DRAW-Paired results after the author’s post-inference exclusion of 50 pairs. Relative to grounded self-review, DeGro improves FDA by 5.8 points for 20B, 8.8 for 120B, and 10.2 for Nano. All three improve abstention and supported repair; 20B RSR rises from 62.3% to 66.7%. DeGro and verdict-only repair tie at 77.8% FDA for 120B: verdict-only repair has higher RSR, while DeGro has higher CAR. Exact-span provenance does not establish semantic entailment, and high UR requires reporting repair utility alongside abstention. Because exclusions followed access to outcomes, these estimates describe the retained subset and do not establish an independent replication.

GPT-OSS 20B improves by +5.83 FDA points (pair-cluster 95% CI [2.5, 9.0]; exact McNemar $p = .00049$; Holm-adjusted $p = .00049$), with 66 versus 31 discordant cases favoring DeGro versus grounded self-review; GPT-OSS 120B improves by +8.83 FDA points (pair-cluster 95% CI [5.7, 12.2]; exact McNemar $p = 6.07 \times 10^{-8}$; Holm-adjusted $p = 1.82 \times 10^{-7}$), with 75 versus 22 discordant cases favoring DeGro versus grounded self-review; Nano improves by +10.17 FDA points (pair-cluster 95% CI [6.0, 14.5]; exact McNemar $p = 1.72 \times 10^{-6}$; Holm-adjusted $p = 3.44 \times 10^{-6}$), with 111 versus 50 discordant cases favoring DeGro versus grounded self-review. These tests use 10,000 pair-cluster bootstrap resamples and Holm adjustment across three model-specific comparisons. All three retained-subset comparisons are significant after adjustment, but exclusions occurred after inference and access to outcomes. The

Query and current formalization			
Source: $-2a + 3b - 4c = 15$, $-a + 2b + 4c = -8$. Find the integer value of a . Encoded constraints: the two displayed equations; all variables are already integers.			Solver verdict: AMBIGUOUS.
Self-Review	Grounded Self-Review	Determinacy-Guided	DeGro (ours)
Model output ADD_CONSTRAINT Cites: "Find the integer value of a ." $a = \text{int}(a)$ Redundant with the declared domain; the target remains ambiguous. ✗	Model output ADD_CONSTRAINT Cites: "integer value of a " $a = \text{int}(a)$ Lexically grounded, but non-resolving and already enforced. ✗	Model output ADD_CONSTRAINT Cites: "Find the integer value of a ." $a \bmod 1 = 0$ Reacts to ambiguity, but invents no missing source relation. ✗	Verified decision ABSTAIN Ambiguous target; no unused source relation supports a repair. No constraint added Correctly withholds an unsupported completion. ✓

Figure 4: Illustrative GPT-OSS 120B output on MIRA-Math case 1in-000271. The source supplies two equations for three integer variables, so the target is not uniquely determined. All three baselines add an integrality condition that is already enforced and does not resolve ambiguity; DeGro is the only method to abstain. This selected example explains the mechanism and is not a substitute for the aggregate comparison in Table 2.

post-hoc results do not establish a preregistered replication or a model-universal benefit.

5 Analysis

5.1 Additional and Natural-Error Analysis

Held-out comparison. The supplement retains the 120-pair results separately: DeGro minus grounded self-review is +5.0 FDA points for 20B (CI [0.4, 9.6]), +6.25 for 120B (CI [1.7, 10.8]), and +1.25 for Nano (CI [-2.5, 5.0]). The direction agrees with the 300-pair results, but neither Nano estimate establishes an incremental FDA benefit. The main comparison does not replace an independent held-out evaluation.

Exploratory source-coverage prompt. Error analysis on the 80-pair development set motivated one frozen prompt refinement: explicitly compare every source condition with F , distinguish ambiguity of F from missing information in the source, and expose already enforced variable domains. The 300 solver-compatible MIRA pairs were already exhausted, so we tested this refinement on a new 100-pair synthetic stress set spanning four algebraic families; this is neither an external benchmark nor an additional MIRA test set. We concurrently reran the original DeGro prompt. For Nano 30B, FDA rises from 75.5% to 95.0% (+19.5 points, pair-cluster 95% CI [13.5, 25.5]; exact $p < .001$), with RSR rising from 52.0% to 90.0% and CAR from 99.0% to 100.0%. This increased repair activity also raises UR from 1/53 (1.9%) to 8/98 (8.2%). For GPT-OSS 120B, FDA changes from 98.5% to 99.5% (+1.0 point, CI [-1.0, 3.0]; $p = .625$), while UR falls from 3/102 (2.9%) to 1/101 (1.0%). Holm adjustment over these two exploratory tests gives $p < .001$ and $p = .625$. The supplement reports the full protocol and table; these results motivate a future frozen external evaluation and do not revise the main comparisons above.

RQ3: human-verified natural errors. On the supported MATH-500 cohort, 158 of 292 formalizations are correct; the remaining labels are 69 wrong-target, 32 wrong-constraint, 24 target-critical missing-constraint, eight extra-constraint, and one wrong-domain. The detector identifies 21 of 24 target-critical omissions (87.5% recall, 95% Wilson CI [69.0, 95.7]) but precision is 17.4% (21/121), and only one omission is successfully repaired end to end (1/24 = 4.2%). DeGro nevertheless raises cohort answer accuracy from 49.3% (144/292) to 53.8% (157/292); 16 discordant cases favor DeGro and three favor the structured baseline (exact McNemar $p = .0044$). This prevalence is conditional on 58.4% solver-fragment coverage.

ALG514 [34] provides a higher-coverage contrast: 503 of 514 cases enter the verified cohort, of which 477 are correct, 11 wrong-target, eight missing-constraint, and seven wrong-constraint. The semantic-error rate is 5.2% (95% Wilson CI [3.6, 7.5]). Structured Solver and DeGro score 94.2% (484/514) and 94.4% (485/514), respectively, using order-invariant subset matching with 10^{-3} tolerance for ALG514’s all-unknown, rounded solution vectors. Nine numerically compatible answers retain a human-verified semantic error, directly showing that answer-only evaluation can conceal representation failures. The one-case DeGro gain on ALG514 is exploratory.

RQ3: detection-repair decomposition. The RQ3 results separate three stages that an aggregate answer score conflates. First, representation coverage asks whether the structured language and solver can express and decide the generated model. Second, ambiguity detection asks whether target-divergent assignments exist within that model. Third, grounded repair asks whether the original source supports a new constraint that removes the disagreement. MATH-500 performs moderately on the first stage (292 of 500), strongly on recall at the second (21 of 24 omissions), and poorly on the third (one of 24). Thus RQ3 identifies conversion of a solver counterexample into a semantically correct, cited edit as the main bottleneck on this cohort.

Precision provides the complementary view. Only 21 of 121 ambiguous MATH-500 specifications are verified target-critical omissions. The remaining ambiguous cases include wrong targets, wrong constraints, and representations whose feasible set is broad for reasons other than a missing source fact. These are legitimate warnings about F but not opportunities for additive repair. A high-recall detector can consequently be useful for routing while still requiring a strict gate: automatically treating every ambiguity as an omission would reproduce the over-repair behavior observed in generic self-review.

Answer accuracy alone also understates the semantic distinction. On MATH-500, DeGro changes the shared structured baseline from 49.3% to 53.8%, with 16 discordances favoring DeGro and three favoring the baseline. Yet only one verified omission is repaired end to end, so most accuracy changes do not constitute evidence that the missing-relation mechanism was solved. On ALG514, nine numerically compatible outputs still carry a verified semantic defect. We therefore report numerical, semantic, functional, grounding, and determinacy outcomes separately rather than reinterpret every answer improvement as representation repair.

Decision-theoretic reading. FDA assigns equal mass to repair on complete sources and abstention on underspecified sources because every controlled item contributes one of each. Real applications may penalize fabrication more heavily than missed repair. The case-level release permits alternative utilities, but the qualitative tradeoff is already visible: generic grounded review attains high RSR while over-repairing, whereas a bare determinacy verdict favors abstention. DeGro attempts to recover useful repairs without surrendering the conservative action boundary. Whether that balance is preferable depends on the downstream cost of an unsupported assertion, which should be declared rather than hidden inside a single accuracy metric.

5.2 Witness Ablation and Family Effects

RQ3: witness ablation. On the exploratory 100-pair set, DeGro scores 98.5 FDA versus 97.5 with witnesses (five versus three discordances favoring DeGro; Holm-adjusted $p = .727$). A minimal witness raises RSR by six points over a bare verdict, also non-significantly ($p = .714$). Concrete assignments therefore add no measurable benefit here: the typed verdict can be exposed by default and witnesses retained for audit or harder cases.

This part of RQ3 is answered only on the exploratory controlled set. Every instance contains one target-critical omission, so non-uniqueness already localizes the relevant defect; witness assignments add tokens but little new information. Specifications with several missing facts or many nuisance dimensions may differ, but these data support determinacy-guided grounding, not a universal benefit from exposing concrete assignments.

RQ1: unsupported additions and repair tradeoffs. For 20B, a bare verdict raises CAR to 98.0% while lowering RSR to 87.7%; DeGro restores repairs, reaches 99.0% CAR, and reduces unsupported additions from 12.5% to 3.4% relative to grounded self-review. For GPT-OSS 120B, DeGro raises CAR from 79.7% to 93.3% and RSR from 97.0% to 98.0%, while reducing unsupported additions from

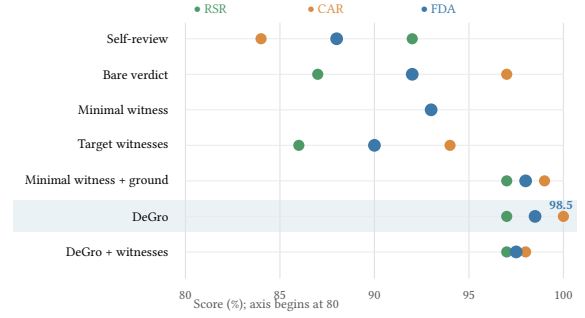


Figure 5: Exploratory GPT-OSS 20B ablation on 100 development pairs. Concrete witnesses do not improve FDA over DeGro without witnesses. ORR is omitted because it is exactly 100 – CAR.

17.3% to 7.0%. Family effects remain concentrated rather than uniform.

Figure 6 explains the RQ1 tradeoff behind the aggregate scores. Generic review often inserts a plausible but unstated relation; DeGro blocks many such additions. The 20B rank-deficient family remains an adverse regime, whereas 120B improves in all five families. Lower UR therefore cannot by itself establish better grounded decision-making; RSR, CAR, and family-specific losses must be read with it.

5.3 Coverage Analysis

RQ3: broad-benchmark coverage audit. To measure the representation bottleneck preceding verification, GPT-OSS 20B generated one shared Structured SymCode specification for each MATH-500 problem [19]. Of 500 specifications, 292 (58.4%) parsed, reached the solver, and produced a conclusive verdict; Figure 7 accounts for every exclusion.

Coverage is fixed solely from parsing and solver state, not observed answers. Solver-level unsupported cases dominate and require a broader theory or compiler; model-level failures instead require better structured generation. Conditional accuracy concerns the 292 supported problems, whereas end-to-end accuracy assigns failures to the other 208. All Structured Solver records completed; DeGro completed 285, with six call errors and one unsupported record. We retain the frozen denominator and count all seven as incorrect.

This separation prevents a favorable post-hoc coverage slice. Conditional accuracy asks how well the agent performs once a specification reaches a conclusive solver state; end-to-end accuracy additionally charges every representation and execution failure. Reporting only the former would make a narrow verifier appear competitive with unrestricted program generation while hiding 41.6% of the benchmark. The exclusion categories also route improvements to the right component: unsupported solver constructs call for broader compilation, while failures before a valid MODEL-SPEC call for better structured generation.

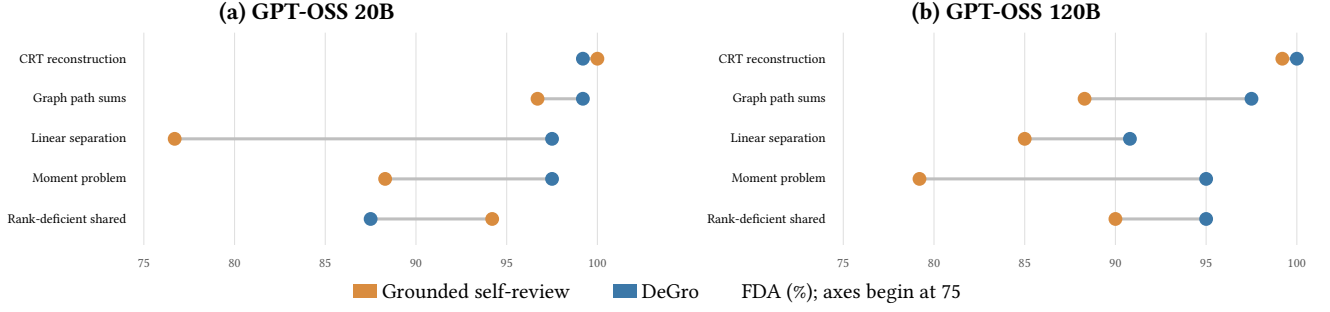


Figure 6: MIRA-Math FDA by family; each family contains 60 pairs. DeGro improves four of five families for GPT-OSS 20B and all five for GPT-OSS 120B. The 20B rank-deficient family is the sole adverse regime.

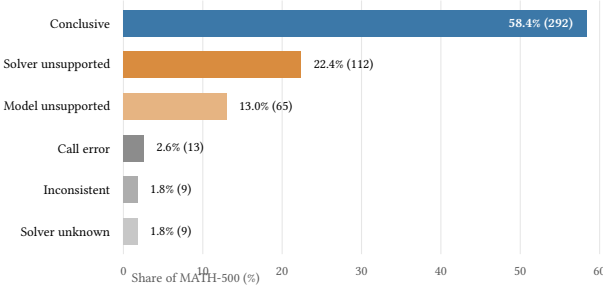


Figure 7: Frozen MATH-500 cohort accounting for all 500 shared initial structured specifications. Solver-level unsupported cases are the largest exclusion category.

5.4 Discussion and Limitations

Interpretation and validity. The paired design holds latent values, target, domains, retained constraints, and solver verdict fixed; only the source support for c^* changes. The repair policy never receives c^* , the gold target, or the gold action. The 120 held-out pairs were frozen before inspection. The 300-pair main evaluation includes pilot and development observations used during method development, so it is not an independent held-out test. The witness ablation and source-coverage refinement are exploratory. The latter was designed from development errors and tested only on synthetic instances from the same task templates; its large Nano gain requires confirmation on an independently sourced frozen benchmark. FDA equally weights repair and correct abstention; UR must be interpreted with RSR and CAR because always abstaining produces zero unsupported additions but no repair utility.

A conventional mixture of complete and incomplete questions admits shortcuts: families may differ in difficulty, wording may reveal that a fact is missing, and a policy may simply learn to repair every ambiguous solver state. Our paired construction rules out these shortcuts because both members share the same formalization and diagnostic. Correct behavior changes only with the natural-language support for the withheld constraint. Exact McNemar tests then depend only on pairs for which the two policies disagree, while pair-cluster bootstrap intervals preserve the dependence between the two case types.

The primary contrast also controls the language-model interface. Both methods receive the same source, specification, output schema, grounding instruction, model, and token budget; the added determinacy signal is the intended treatment. Gold latent values are used only after generation for scoring. A direct target pin is considered unsupported unless a source span licenses it and all semantic, functional, and determinacy gates accept it. Retaining raw and gated outputs reveals when apparent accuracy is purchased by fabrication.

Scope of the claim. The evidence supports target determinacy as a guard against plausible but unstated repairs, including a five-point GPT-OSS gain. It does not establish semantic equivalence between P and F , a universal grounding benefit, an incremental witness benefit, or superiority over unrestricted code generation. Exact spans are auditable provenance rather than semantic entailment. Confidence and determinacy are complementary: low confidence may justify fallback for a determinate target, while confidence cannot authorize an ambiguous one.

The distinction is important for agent design. An execution monitor answers whether an artifact runs; an outcome verifier may check a final answer; DeGro instead characterizes the feasible set before the agent commits to a value. This intermediate certificate supports selective action: answer when the target is determined, repair only when a cited source fact restores determinacy, and otherwise ask, abstain, or escalate. It is not a proof that the formalization matches the source. A correctly solved but incorrectly translated model may remain determinate.

Grounding itself is also only a proxy. A model can cite the correct sentence yet encode the wrong equation, while an implicit mathematical convention may justify an expression without a contiguous span. We therefore keep semantic validity, functional validity, provenance, and solver determinacy as separate fields. This decomposition makes disagreement auditable and prevents numerical answer correctness from silently absorbing translation errors.

External validity. MIRA cases are controlled but templated; MATH-500 is natural but only 58.4% lies inside the present solver fragment, whereas ALG514 reaches 97.9% coverage in a narrower domain. Labels were human-verified after model proposals, not independently annotated, and dataset prevalence is therefore reported separately. Natural errors contain only 24 MATH-500 and eight ALG514 target-critical omissions: detection is high but repair

remains limited. The two GPT-OSS sizes also differ materially in baseline abstention behavior, bounding any size-general claim.

The controlled benchmark deliberately removes one known fact and therefore estimates a clean repair-versus-abstain decision under idealized representation control. It underrepresents paraphrase, irrelevant prose, diagrams, implicit conventions, wrong constraints, and interacting translation errors. The natural cohorts contain these phenomena but trade away causal control. Their purposes are complementary: the paired benchmark identifies a policy effect, while the natural study estimates what the current pipeline actually encounters. We do not merge them into one leaderboard number.

The annotation protocol also avoids defining positives with DeGro’s own verdict. Detector outputs, repairs, method names, and downstream answer outcomes are hidden from the initial packet. A target-critical omission must instead be established from the source and generated specification; correct, wrong-constraint, wrong-target, and wrong-domain cases remain in the denominator. Because model proposals seeded review and no separate blinded pre-discussion labels were preserved, we claim human verification rather than independent annotation or inter-annotator reliability.

Reproducibility and risk. The artifact includes the compiler, two-copy query, gates, tests, builders, frozen manifests, prompts, raw records, and analysis, including both additional Nemotron runs. Each MIRA-Math main analysis consumes 2,400 records per model; the supplement retains the 120-pair outputs and their complete hashes. Determinacy remains relative to F : wrong or extra constraints may force a unique wrong target, non-target omissions are invisible, and the current SMT fragment excludes much nonlinear, transcendental, and diagrammatic reasoning. Interfaces must expose this scope rather than imply truth. The data contain no personal information; AI assistance requires venue-compliant disclosure and does not reduce author responsibility.

The manifests record dataset revision and license, selection seed, family counts, exact model strings, decoding settings, prompt hashes, timestamps, and full output hashes. Analysis scripts reconstruct every table from JSONL, reject incomplete expected pair counts, and preserve whether a row came from pilot, development, or confirmatory data. Natural-error packets and verified labels are separately blinded and content-addressed. These records reproduce the 300-pair main table and the separate 120-pair held-out comparison.

Several failure modes remain outside the certificate. A wrong or extra constraint may force a unique but incorrect target; an omitted fact that does not affect q is invisible; several interacting omissions may exceed the bounded repair budget; and witnesses may differ only on irrelevant variables. The supported SMT fragment excludes much nonlinear, transcendental, and diagrammatic mathematics. These are reasons to surface the encoded constraints, cited repair, and solver status in the interface, not to display an unqualified “verified” badge.

Finally, the tested models exhibit different instruction-following and abstention behavior, and hosted checkpoints may change. The additional Nano result shows that strong baseline abstention can leave little incremental FDA benefit for DeGro. DeGro’s advantage should therefore be treated as a property of a specified model-prompt-gate configuration. The GPT-OSS 120B baseline

over-repairs many underspecified cases despite its larger size. Deterministic gates remain necessary, and deployment thresholds should reflect the relative cost of fabricated facts and missed repairs.

Deployment implications. DeGro is best used as a routing signal rather than a universal correctness badge. A determinate, supported specification can proceed to answer generation; an ambiguous specification with a grounded candidate can enter the bounded repair loop; and an ambiguous specification without support should trigger clarification or abstention. Inconsistent, unknown, and unsupported cases require different fallbacks, as Table 1 makes explicit. Keeping these states typed prevents an agent from treating lack of a solver certificate as permission to guess.

The results also suggest a conservative interface. The system should expose the target expression, the relevant encoded constraints, the determinacy status, and any source span used for repair. When a witness exists, it can be retained in the audit record even if the default language-model prompt receives only the verdict. This design follows the ablation result: witnesses are valuable mechanical evidence, but displaying them to the repair model did not improve decisions on the present benchmark. Users can then distinguish “the current model fixes this quantity” from the stronger and unsupported claim that the entire translation is correct.

Deployment comparisons should report representation coverage, decision quality conditional on a conclusive verifier state, and end-to-end utility with unsupported, unknown, and call-error cases retained as failures. Otherwise, a narrow symbolic verifier can appear reliable by declining difficult inputs before scoring.

6 Conclusion

DRAW-Paired extends the design to 300 retained author-reviewed natural algebra sources. DeGro improves FDA over grounded self-review across the three models, and ties verdict-only repair for GPT-OSS 120B. Both cohorts were reviewed after inference, the extension was selected after initial outcomes, and 50 pairs were subsequently excluded after re-review; these post-hoc results complement the controlled MIRA-Math evaluation. On 300 MIRA-Math pairs, DeGro raises FDA over grounded self-review by 5.0 points for GPT-OSS 20B and 7.33 for 120B, with lower unsupported additions for both; 20B sacrifices some supported repairs. Nano’s +0.83-point difference is inconclusive, limiting model-general claims. The main set includes pilot and development observations; separate 120-pair held-out results remain in the supplement. For RQ2, decisions depend on source support even when formalization and determinacy signal are identical. For RQ3, coverage and detection substantially exceed end-to-end grounded repair on natural errors, while concrete witnesses add no measurable benefit on the exploratory single-omission set. Target determinacy is a representation-relative routing condition, with grounding and coverage reported alongside decision accuracy.

References

- [1] Charbel Al Bateh and Samer Saab Jr. 2026. MIRA-Math: A Benchmark for Minimal Information Requesting and Mathematical Reasoning. *arXiv preprint arXiv:2607.07391* (2026).
- [2] Mohammad Alhanahnah, Md Rashedul Hasan, and Hamid Bagheri. 2024. An Empirical Evaluation of Pre-trained Large Language Models for Repairing Declarative Formal Specifications. *arXiv preprint arXiv:2404.11050* (2024).
- [3] Paul C. Attie, Kinan Dak Al Bab, and Mouhammad Sakr. 2017. Model and Program Repair via SAT Solving. *ACM Transactions on Embedded Computing Systems* 17, 2, Article 32 (2017). doi:10.1145/3147426
- [4] Zhangir Azerbayev, Bartosz Piotrowski, Hailey Schoelkopf, Edward W. Ayers, Dragomir Radev, and Jeremy Avigad. 2023. ProofNet: Autoformalizing and Formally Proving Undergraduate-Level Mathematics. *arXiv preprint arXiv:2302.12433* (2023).
- [5] Sina Bagheri Nezhad, Yao Li, and Ameeta Agrawal. 2026. SymCode: A Neurosymbolic Approach to Mathematical Reasoning via Verifiable Code Generation. In *Findings of the Association for Computational Linguistics: EACL 2026*. Association for Computational Linguistics, 1489–1503. doi:10.18653/v1/2026.findings-eacl.76
- [6] Gilles Barthe, Juan Manuel Crespo, and César Kunz. 2011. Relational Verification Using Product Programs. In *FM 2011: Formal Methods (Lecture Notes in Computer Science, Vol. 6664)*. 200–214. doi:10.1007/978-3-642-21437-0_17
- [7] Gilles Barthe, Pedro R. D’Argenio, and Tamara Rezk. 2004. Secure Information Flow by Self-Composition. In *17th IEEE Computer Security Foundations Workshop*. 100–114. doi:10.1109/CSFW.2004.1310735
- [8] Wenhui Chen, Xueguang Ma, Xinyi Wang, and William W. Cohen. 2023. Program of Thoughts Prompting: Disentangling Computation from Reasoning for Numerical Reasoning Tasks. *Transactions on Machine Learning Research* (2023).
- [9] Edmund M. Clarke, Orna Grumberg, Suresh Jha, Yuan Lu, and Helmut Veith. 2000. Counterexample-Guided Abstraction Refinement. In *Computer Aided Verification (Lecture Notes in Computer Science, Vol. 1855)*. 154–169. doi:10.1007/10722167_15
- [10] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training Verifiers to Solve Math Word Problems. In *arXiv preprint arXiv:2110.14168*.
- [11] Wenyao Cui, Huaping Zhang, Yongyi Huang, Qiuchi Li, Jian Xu, Cheng-Lin Liu, Chunxiao Gao, Juan Wang, and Baohua Zhang. 2026. SymDiag: Explainable Diagnosis for LLM Reasoning via Neuro-Symbolic Verification. *arXiv preprint arXiv:2608.08786* (2026).
- [12] Leonardo de Moura and Nikolaj Bjørner. 2008. Z3: An Efficient SMT Solver. In *Tools and Algorithms for the Construction and Analysis of Systems (Lecture Notes in Computer Science, Vol. 4963)*. Springer, 337–340. doi:10.1007/978-3-540-78800-3_24
- [13] Isil Dillig, Thomas Dillig, Boyang Li, and Kenneth L. McMillan. 2013. Inductive Invariant Generation via Abductive Inference. In *Proceedings of the 34th ACM SIGPLAN Conference on Programming Language Design and Implementation*. 443–456. doi:10.1145/2491956.2462187
- [14] Ran El-Yaniv and Yair Wiener. 2010. On the Foundations of Noise-Free Selective Classification. *Journal of Machine Learning Research* 11 (2010), 1605–1641.
- [15] Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. PAL: Program-aided Language Models. In *International Conference on Machine Learning*.
- [16] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems*, Vol. 30.
- [17] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujia Yang, Nan Duan, and Weizhu Chen. 2024. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing. In *International Conference on Learning Representations*.
- [18] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujia Yang, Minlie Huang, Nan Duan, and Weizhu Chen. 2024. ToRA: A Tool-Integrated Reasoning Agent for Mathematical Problem Solving. In *International Conference on Learning Representations*.
- [19] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring Mathematical Problem Solving With the MATH Dataset. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, Vol. 1.
- [20] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2024. Large Language Models Cannot Self-Correct Reasoning Yet. In *International Conference on Learning Representations*.
- [21] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. In *International Conference on Learning Representations*.
- [22] Junbo Jacob Lian, Yujun Sun, Huiling Chen, Chaoyu Zhang, Hanzhang Qin, and Chung-Piaw Teo. 2026. ReLoop: Structured Modeling and Behavioral Verification for Reliable LLM-Based Optimization. *arXiv preprint arXiv:2602.15983* (2026).
- [23] Mark H. Liffiton and Karem A. Sakallah. 2008. Algorithms for Computing Minimal Unsatisfiable Subsets of Constraints. *Journal of Automated Reasoning* 40 (2008), 1–33. doi:10.1007/s10817-007-9084-z
- [24] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2023. Let’s Verify Step by Step. *arXiv preprint arXiv:2305.20050* (2023).
- [25] Qing Lyu, Shreya Havaldar, Adam Stein, Li Zhang, Delip Rao, Eric Wong, Marianna Apidianaki, and Chris Callison-Burch. 2023. Faithful Chain-of-Thought Reasoning. In *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*. 305–329. doi:10.18653/v1/2023.ijcnlp-main.20
- [26] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems*, Vol. 36.
- [27] NVIDIA. 2025. Nemotron 3 Nano: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning. *arXiv:2512.20848*. doi:10.48550/arXiv.2512.20848
- [28] OpenAI. 2025. gpt-oss-120b & gpt-oss-20b Model Card. Accessed 2026-09-09. <https://openai.com/index/gpt-oss-model-card/>
- [29] Raymond Reiter. 1987. A Theory of Diagnosis from First Principles. *Artificial Intelligence* 32, 1 (1987), 57–95. doi:10.1016/0004-3702(87)90062-2
- [30] Dipankar Sarkar. 2026. From Errors to Proofs: Minimal-Core-Guided Repair for Neuro-Symbolic Constraint Solving. *arXiv preprint arXiv:2608.14771* (2026).
- [31] Armando Solar-Lezama, Liviu Tancau, Rastislav Bodik, Sanjit A. Seshia, and Vijay A. Saraswat. 2006. Combinatorial Sketching for Finite Programs. In *Proceedings of the 12th International Conference on Architectural Support for Programming Languages and Operating Systems*. 404–415. doi:10.1145/1168857.1168907
- [32] Tachio Terauchi and Alex Aiken. 2005. Secure Information Flow as a Safety Problem. In *Static Analysis (Lecture Notes in Computer Science, Vol. 3672)*. 352–367.
- [33] Jonathan Uesato, Nate Kushman, Ramana Kumar, Francis Song, Noah Siegel, Lisa Wang, Antonia Creswell, Geoffrey Irving, and Irina Higgins. 2022. Solving Math Word Problems with Process- and Outcome-Based Feedback. *arXiv preprint arXiv:2211.14275* (2022).
- [34] Shyam Upadhyay and Ming-Wei Chang. 2017. Annotating Derivations: A New Evaluation Strategy and Dataset for Algebra Word Problems. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers*. Association for Computational Linguistics, 494–504. <https://aclanthology.org/E17-1047/>
- [35] Ke Wang, Houxing Ren, Aojun Zhou, Zimu Lu, Sichun Luo, Weikang Shi, Renrui Zhang, Linqi Song, Mingjie Zhan, and Hongsheng Li. 2024. MathCoder: Seamless Code Integration in LLMs for Enhanced Mathematical Reasoning. In *International Conference on Learning Representations*.
- [36] Yuhuai Wu, Albert Qiaoqiang Jiang, Wenda Li, Markus N. Rabe, Charles Staats, Mateja Jamnik, and Christian Szegedy. 2022. Autoformalization with Large Language Models. In *Advances in Neural Information Processing Systems*, Vol. 35.